Annex: 3. Constraint-Based Problem Solving

1. DNS Resolution – Google

Open the website:

Google: https://www.google.com/?zx=1787560655258

Start Wireshark before opening the browser.

Constraint

You are allowed to inspect only DNS packets.

Task: Identify the packet containing the DNS response that resolves www.google.com.

Wireshark Filter: dns

Correct Packet Number: 489

Observation: Packet 489 is the DNS response for www.google.com and contains the returned IPv4 address records.

dns

No.	Time	Source	Destination	Protocol	Lengtl	Info
336	7.193289100	10.133.185.43	10.133.185.81	DNS	81	Standard query 0xf2c7 AAAA update.googleapis.com
337	7.193592100	10.133.185.43	10.133.185.81	DNS	81	Standard query 0xa4cd A update.googleapis.com
338	7.200747900	10.133.185.81	10.133.185.43	DNS	109	Standard query response 0xf2c7 AAAA update.googleapis.com AAAA 2404:6800:4007:807::200e
339	7.200748900	10.133.185.81	10.133.185.43	DNS	97	Standard query response 0xa4cd A update.googleapis.com A 142.251.222.142
344	7.220648900	10.133.185.81	10.133.185.43	DNS	146	Standard query response 0x98ef HTTPS clientservices.googleapis.com SOA ns1.google.com
346	7.351602400	10.133.185.81	10.133.185.43	DNS	105	Standard query response 0xe3f9 A clientservices.googleapis.com A 172.217.24.142
347	7.351603100	10.133.185.81	10.133.185.43	DNS	138	Standard query response 0x2987 HTTPS update.googleapis.com SOA ns1.google.com
477	9.694778700	10.133.185.43	10.133.185.81	DNS	76	Standard query 0x8f9d AAAA mtalk.google.com
478	9.695289000	10.133.185.43	10.133.185.81	DNS	76	Standard query 0x7005 A mtalk.google.com
484	9.805827000	10.133.185.43	10.133.185.81	DNS	74	Standard query 0xe6d7 HTTPS www.google.com
485	9.806469100	10.133.185.43	10.133.185.81	DNS	74	Standard query 0xf5c6 AAAA www.google.com
486	9.806875800	10.133.185.43	10.133.185.81	DNS	74	Standard query 0xedf8 A www.google.com
487	9.808485800	10.133.185.81	10.133.185.43	DNS	133	Standard query response 0x8f9d AAAA mtalk.google.com CNAME mobile-gtalk.l.google.com AAAA 2404:6800:4000:1025::bc
488	9.918318100	10.133.185.81	10.133.185.43	DNS	298	Standard query response 0xf5c6 AAAA www.google.com AAAA 2001:4860:482d:7700:: AAAA 2001:4860:4826:7700:: AAAA 20…
489	9.918319800	10.133.185.81	10.133.185.43	DNS	202	Standard query response 0xedf8 A www.google.com A 142.251.150.119 A 142.251.154.119 A 142.251.157.119 A 142.251…

2. HTTPS Web Access – Wikipedia

Clear the browser cache, start Wireshark, and open:

Wikipedia: https://www.wikipedia.org/

Constraint

You may use only one display filter and cannot inspect browser developer tools.

Task: Identify the transport-layer protocol carrying the encrypted web connection.

Correct Answer: TCP
Wireshark display filter used: tcp

8065	133.710932400	2606:4700:83b1:d8b7…	2401:4900:9245:faf9…	TLSv1.2	98	Application Data
8066	133.711371900	2401:4900:9245:faf9…	2606:4700:83b1:d8b7…	TLSv1.2	102	Application Data
8067	133.827033600	2606:4700:83b1:d8b7…	2401:4900:9245:faf9…	TCP	74	443 → 51979 [ACK] Seq=1325 Ack=1277 Win=17 Len=0
8068	133.827371400	2a03:2880:f368:120:…	2401:4900:9245:faf9…	TCP	74	[TCP Dup ACK 8055#1] 5222 → 60487 [ACK] Seq=489 Ack=1335 Win=2062 Len=0
8070	133.834476200	2a03:90c0:321:2803:…	2401:4900:9245:faf9…	TCP	1185	80 → 60928 [ACK] Seq=852075 Ack=438 Win=8 Len=1099 TSval=347877857 TSecr=910411460
8071	133.834478100	2a03:90c0:321:2803:…	2401:4900:9245:faf9…	TCP	1006	80 → 62493 [PSH, ACK] Seq=503801 Ack=438 Win=32768 Len=920 TSval=131033641 TSecr=910411460
8072	133.834479100	2a03:90c0:321:2803:…	2401:4900:9245:faf9…	TCP	1185	80 → 60928 [ACK] Seq=853174 Ack=438 Win=8 Len=1099 TSval=347877873 TSecr=910411608
8073	133.834639000	2401:4900:9245:faf9…	2a03:90c0:321:2803:…	TCP	86	60928 → 80 [ACK] Seq=438 Ack=854273 Win=10 Len=0 TSval=910411832 TSecr=347877857
8074	133.834771300	2a03:90c0:321:2803:…	2401:4900:9245:faf9…	TCP	1006	80 → 62493 [ACK] Seq=504721 Ack=438 Win=32768 Len=920 TSval=131033684 TSecr=910411661
8075	133.834813700	2401:4900:9245:faf9…	2a03:90c0:321:2803:…	TCP	86	62493 → 80 [ACK] Seq=438 Ack=505641 Win=2560 Len=0 TSval=910411833 TSecr=131033641
8076	133.851091200	10.133.185.43	35.175.89.16	TCP	1414	[TCP Retransmission] 61498 → 443 [PSH, ACK] Seq=29715 Ack=6711 Win=251 Len=1360 [TCP PDU reassembled in 80
8077	133.935833000	2a03:90c0:321:2803:…	2401:4900:9245:faf9…	TCP	1185	80 → 60928 [ACK] Seq=854273 Ack=438 Win=8 Len=1099 TSval=347877982 TSecr=910411832
8078	133.935834900	2a03:90c0:321:2803:…	2401:4900:9245:faf9…	TCP	1144	80 → 60928 [ACK] Seq=855372 Ack=438 Win=8 Len=1058 TSval=347877998 TSecr=910411832
8079	133.935986600	2401:4900:9245:faf9…	2a03:90c0:321:2803:…	TCP	86	60928 → 80 [ACK] Seq=438 Ack=856430 Win=10 Len=0 TSval=910411934 TSecr=347877982
8080	134.032783400	2a03:90c0:321:2803:…	2401:4900:9245:faf9…	TCP	1006	80 → 62493 [ACK] Seq=505641 Ack=438 Win=32768 Len=920 TSval=131033811 TSecr=910411833

3. FTP Traffic Challenge – Real FTP Test Service

FTP Test Server: https://test.rebex.net/

Constraint: Only one port-based filter was used.

Task: Identify the FTP control connection.

Wireshark Filter: tcp.port == 21

Correct Answer: Port 21

4. SMTP Traffic Identification – Mailtrap Sandbox

SMTP Testing Environment: Mailtrap Email Sandbox

Constraint: Only packets belonging to the configured SMTP connection were inspected.

Task: Identify the application protocol responsible for transmitting the outgoing email.

Correct Answer: SMTP